

Precise Temperature-Gradient Regulated Structure–Activity Relationship of PCN-224 as Bifunctional Nanocarriers

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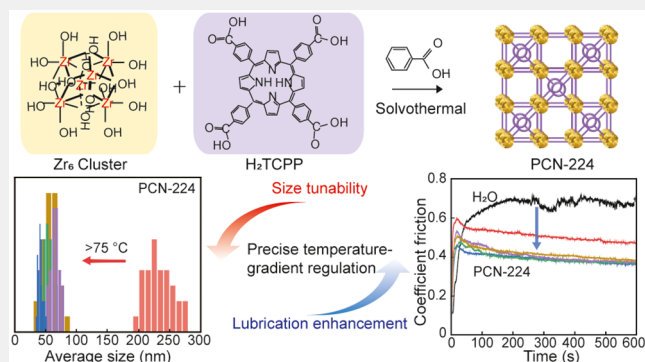
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ABSTRACT: Metal–organic frameworks (MOFs) demonstrate considerable potential for biomedical applications. Due to their high porosity and photoactive porphyrin ligands, PCN-224 is considered an ideal drug delivery nanocarrier. However, the synthesis strategy for tailoring the application of PCN-224 in complex pathological diseases remains unclear. In this study, we investigated the structure–activity relationship of PCN-224 nanoparticles as bifunctional nanocarriers for osteoarthritis (OA) treatment under a precisely controlled temperature gradient. By tuning the temperatures, a series of PCN-224 nanoparticles with adjustable sizes were synthesized and their morphologies transitioned from a spindle-shaped to a spherical-shaped structure. The photoluminescence intensity of PCN-224 nanoparticles decreased as the synthesis temperatures increased. Using PCN-224 nanoparticles as aqueous lubricating additives, both the friction coefficient and wear volume were significantly reduced. Among the MOFs, PCN-224 (synthesized at 105 °C) exhibited excellent antifriction performances and acid-responsive drug release properties, demonstrating their promise as bifunctional OA treatment nanocarriers. Our work enhanced the understanding of the relationships between PCN-224 synthesis conditions and its physicochemical properties relationships, expanding their potential for multifunctional biomedical applications.

KEYWORDS: MOFs, Porphyrin, Photoluminescence, Aqueous lubrication, Drug release



1. INTRODUCTION

Metal–organic frameworks (MOFs), a class of nanoporous materials, are composed of metal ions or clusters linked by multitopic organic ligands, forming extended network structures.^{1,2} Compared with conventional materials, like polymers³ and liposome,⁴ MOFs have emerged as promising drug delivery nanocarriers owing to their high porosity and tunable structures.⁵ For example, Duan and co-workers reported a core–shell structure of ZIF-8-based colloid particles with ultrahigh drug molecule loading up to 90.0%.⁶ A report of Zhang and co-workers has introduced a nucleic acid-functionalized UiO-67 hybrid which can act as sense-and-treat functional drug carriers.⁷ Tuning the metal-nodes/organic-ligands ratio and synthetic parameters, including reaction time, temperature, and pH, directly determines the compatibility between the resulting topology and its functional properties.^{8–10} Reaction temperature has the unique advantage of quantitatively regulating the nucleation rate, ligand-coordination efficiency, and crystal-growth orientation.¹¹ Low temperatures often induce disordered nucleation, whereas high

temperatures may trigger ligand pyrolysis. Therefore, regulating the temperature represents an effective strategy for tailoring MOFs to achieve optimal “uniform morphology + complete crystal structure + adjustable function”.

Among the MOFs families, PCN-224 consist of porphyrin tetracarboxylic acid ligands and zirconium ions nodes, and exhibits unique advantages arising from its rare ordered mesoporous structure.^{12,13} As a representative of Zr-based porphyrin MOFs, PCN-224 is considered an ideal drug delivery candidate owing to its structural tunability, well-defined open channels, exceptional physiological stability, desirable biocompatibility, and inherent photosensitivity, which lay a foundation for multifunctional biomedical

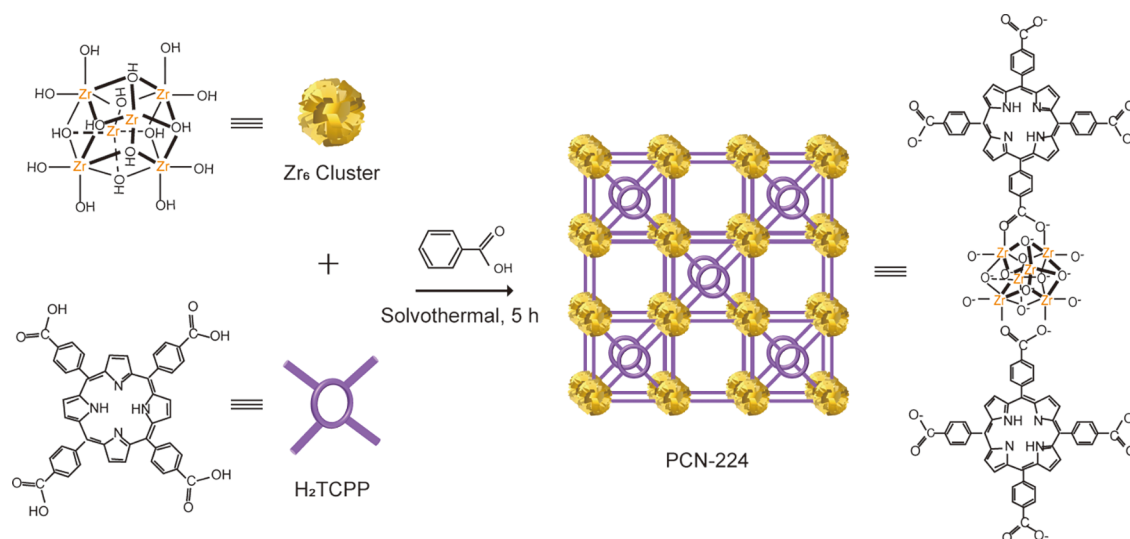
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Scheme 1. Schematic Diagram of PCN-224 Nanoparticles at Different Synthesis Temperatures



applications.^{14,15} For example, Park and co-workers reported PCN-224 nanoparticles with significant PDT efficacy through a bottom-up approach, which can serve as a promising nanoplatform by adopting a small molecule into the tunable framework.¹⁶ Recently, Sun and co-workers developed a PCN-224 nanoplatform for transdermal drug delivery, enabling the nanoparticles and their cargos to penetrate the skin as an intact entity.¹⁷ Although PCN-224 offers such advantages for biomedical applications, challenges such as lack of an in-depth understanding of its synthesis conditions remain in its use for treating diseases with complex pathologies. Moreover, most studies synthesized PCN-224 at a fixed single temperature (usually at 120 °C), and rarely examine how temperature gradients influence its key parameters.^{18,19} Nevertheless, these key parameters fundamentally determine the morphologies and physicochemical properties of MOFs, thereby influencing their drug loading capacity and efficiency.

In clinical practice, some diseases require a customized design of a multifunctional nanoplatform, such as those with imaging or on-demand drug release characteristics, meeting the purpose of nanocarrier tracking and enhanced local efficacy. Osteoarthritis (OA) is a whole-joint and disabling disease that imposes substantial health and socioeconomic burdens on millions of people, particularly for the elderly.^{20,21} Lubrication dysfunction of articular cartilage is considered one of the major causes of OA onset, resulting in the formation of chondral debris and the subsequent heterogenization of chondrocytes and immune cells.²² The chondral debris further facilitates the cellular secretion of proteolytic enzymes, followed by synovial inflammation that further accelerates disease progression.²³ Synergetic therapy is an emerging nanoplatform for OA treatment, integrating long-term joint lubrication and localized drug delivery to minimize the systemic toxicity associated with oral administration.^{24,25} Achieving nanocarriers that simultaneously provide function of both lubrication and drug release remains a central challenge in the development of advanced OA therapies.

Herein, we synthesized a series of PCN-224 nanoparticles under a precisely controlled solvothermal temperature gradient, yielding a potential bifunctional nanocarrier for OA treatment that exhibited both lubricating properties and acid-responsive drug release behavior. Furthermore, we explored

the structure–activity relationship by which solvothermal temperatures regulated the photoluminescence properties of PCN-224 nanoparticles (Scheme 1). A series of PCN-224 nanoparticles were synthesized via solvothermal method under different temperatures ranging from 75 to 135 °C. The nanoparticle size became more uniform and gradually transitioned from a spindle-shaped to spherical-shaped morphology with rising solvothermal synthesis temperatures. The photoluminescence intensity of PCN-224 nanoparticles decreased with increasing synthesis temperatures. When used as an aqueous lubricating additive, PCN-224 produced substantial reductions in both friction coefficient and wear volume. Moreover, the PCN-224 (105 °C) nanocarriers exhibited enhanced antifriction performance and acid-responsive drug release behavior, positioning them as promising bifunctional candidates for synergistic OA therapy.

2. EXPERIMENTAL SECTION

2.1. Materials and Chemicals

5,10,15,20-Tetrakis(4-carboxyphenyl)porphyrin (H₂TCPP) was purchased from Energy Chemical. Zirconyl chloride octahydrate (ZrOCl₂·8H₂O) was purchased from Aladdin Chemical Co. Ltd. *N,N*-Dimethylformamide (DMF), benzoic acid (BA), and acetone were purchased from Shannxi Dexiang Chemical. Diclofenac sodium (DS, 98%) was purchased from Energy Chemical. Deionized water (DI H₂O) was obtained from an ultrapure water purification machine (Ulupure). All commercial chemicals were used without further purification.

2.2. Synthesis of PCN-224 Nanoparticles

H₂TCPP (100 mg, 0.13 mmol), ZrOCl₂·8H₂O (300 mg, 0.93 mmol), and benzoic acid (2.8 g, 23 mmol) were dissolved in 80 mL of DMF in a 250 mL round-bottom flask. The mixture was stirred at 300 rpm and heated via solvothermal temperatures at 75, 90, 105, 120, and 135 °C for 5 h, respectively. Following the reaction, a series of PCN-224 nanoparticles were isolated by centrifugation at 15000 rpm for 30 min and subsequently washed three times with fresh DMF and acetone. Finally, the resulting PCN-224 (75–135 °C) nanoparticles were dispersed in water for further characterization and analysis.

2.3. Synthesis of PCN-224@DS Nanocarriers

A 50 mg amount of PCN-224 was added into a 100 mL single neck flask within 50 mL of DI water. Then, 2.5, 5, 10, 15, and 25 mg of DS were respectively added into the above suspension and stirred at 25 °C overnight. The mixture was isolated by centrifugation at 15000

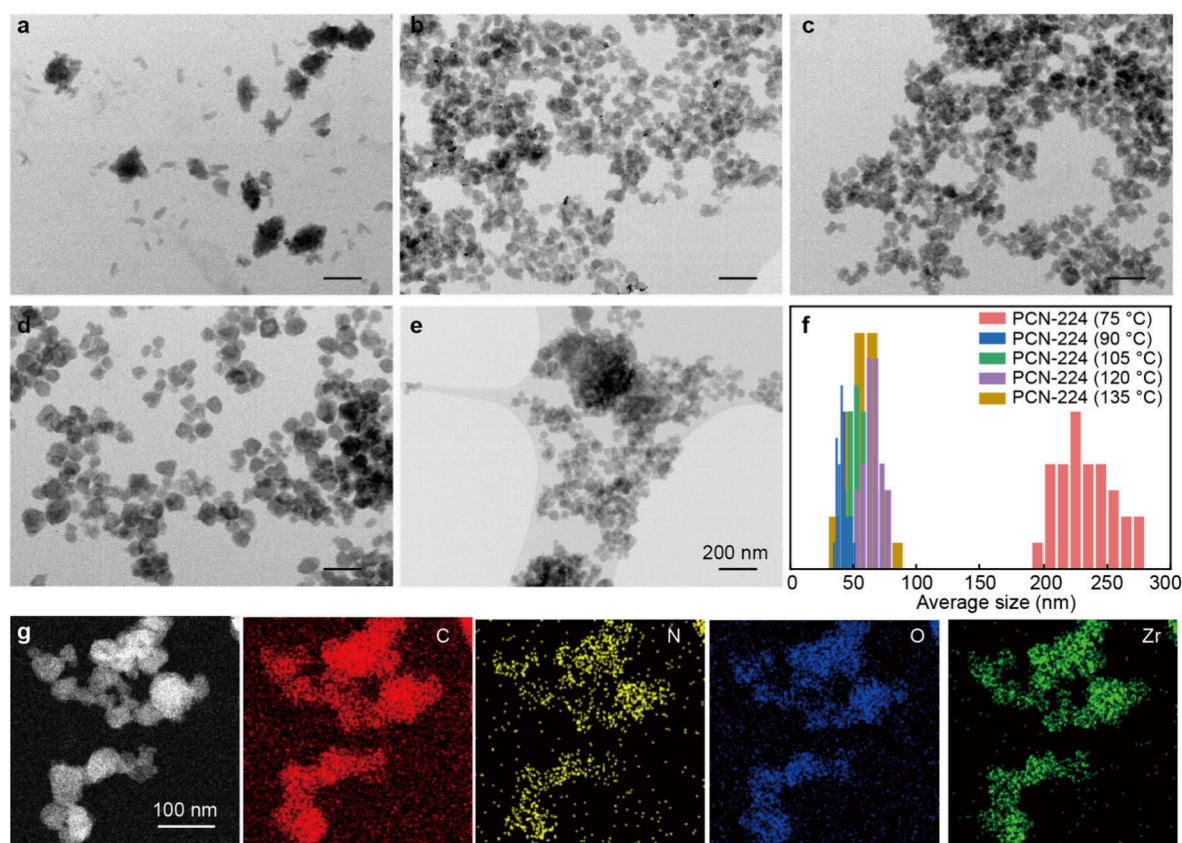


Figure 1. Average size and shape of PCN-224 nanoparticles at different solvothermal synthesis temperatures. TEM images of (a) PCN-224 (75 °C), (b) PCN-224 (90 °C), (c) PCN-224 (105 °C), (d) PCN-224 (120 °C), and (e) PCN-224 (135 °C) nanoparticles. (f) Average size of the above PCN-224 nanoparticles. (g) HAADF image of PCN-224 (105 °C) and the corresponding EDS mapping of C, N, O, and Zr elements.

rpm for 30 min and washed by DI H₂O for two times to remove excess drugs. The centrifugation products were then redispersed in 10 mL of DI H₂O for further use and named PCN-224@DS nanocarrier. The 3 mL supernatant was collected into a centrifuge tube for evaluating the drug loading capability of the nanocarrier.

2.4. Characterizations

Morphologies of a series of PCN-224 (75–135 °C) nanoparticles were measured by transmission electron microscopy (JEOL GrandARM2 TEM) at 300 kV at TU Berlin (Germany). Crystallinity and phase purity were tested from powder X-ray diffraction (PXRD, Bruker D8) using Cu K α radiation ($\lambda = 1.5404$ Å) at a voltage of 40 kV and a current of 40 mA. The chemical structure of the PCN-224 nanoparticles was obtained from Fourier transform infrared spectroscopy (FT-IR, Bruker Tensor II). Thermogravimetric analysis (TG) measurement was tested in a Mettler Toledo apparatus, with a temperature measurement range of 35–800 °C (10 °C/min). Transmittance curves of the PCN-224 nanoparticles were determined by ultraviolet–visible spectroscopy (UV–vis, LAMBDA 365). Photoluminescence spectra were obtained by a FLS 980 spectrofluorometer with a xenon lamp as the light source and with correction against the photomultiplier tube response. Tribological properties and wear scar images of the pure water and the PCN-224 nanoparticles were assessed using ball (6.35 mm)-on-disk contacts and measured on the universal material tester (UMT Tribo) and 3D optical profiler (Bruker, GTK-18-0433).

Drug delivery properties analysis of PCN-224 (105 °C) nanoparticles used a dialysis bags ($M_w = 3500$) diffusion method. The amounts of loaded DS in PCN-224 (105 °C) nanoparticles were determined by the amounts of added drugs and drugs remaining in the supernatant after centrifugation. A 20 mg amount of DS@PCN-224 (105 °C) nanocarriers was placed into dialysis bags within 3 mL of H₂O solution at pH = 5.5 and 7.0 and, then, transferred to jars

containing 27 mL of H₂O at pH = 5.5 and 7.0. The jar was constantly shaken in a shaking incubator under 35 °C. A time-dependent UV–vis spectra was used to measure the absorbance of the collected drugs at 276.5 nm. Based on the standard curve of DS, the drug loading capacity (DLC) and efficiency (DLE) in different solution conditions were evaluated.

3. RESULTS AND DISCUSSION

3.1. Morphology Study of PCN-224 Nanoparticles

Figure 1 depicts the morphology of a series of PCN-224 nanoparticles at different solvothermal synthesis temperatures (75–135 °C) by transmission electron microscopy (TEM) images. In Figure 1a, spindle-shaped PCN-224 (75 °C) nanoparticles and rod-shaped porphyrin molecules can be observed.²⁶ At lower temperatures, the spindle-like feature of PCN-224 (75 °C) nanoparticles originated from the intrinsic rod-shaped structure of porphyrin. As the solvothermal synthesis temperatures increased, the morphology of the PCN-224 nanoparticles gradually transitioned from a spindle-shaped structure to a spherical-shaped structure (Figure 1b–e). Notably, PCN-224 (135 °C) nanoparticles tend to agglomerate at the highest solvothermal synthesis temperature. The average size (calculating the number of 50 nanoparticles) distributions of PCN-224 (75–135 °C) nanoparticles were plotted based on their TEM images (Figure 1f). The analysis of distributions revealed the temperature-dependent variation in the nanoparticle average size of the PCN-224 nanoparticles. As the solvothermal synthesis temperature rose from 75 to 135 °C, the average nanoparticle sizes of PCN-224 were 229.3 ± 25.6 nm (75 °C), 40.9 ± 3.9 nm (90 °C), 56.7 ± 4.2 nm (105

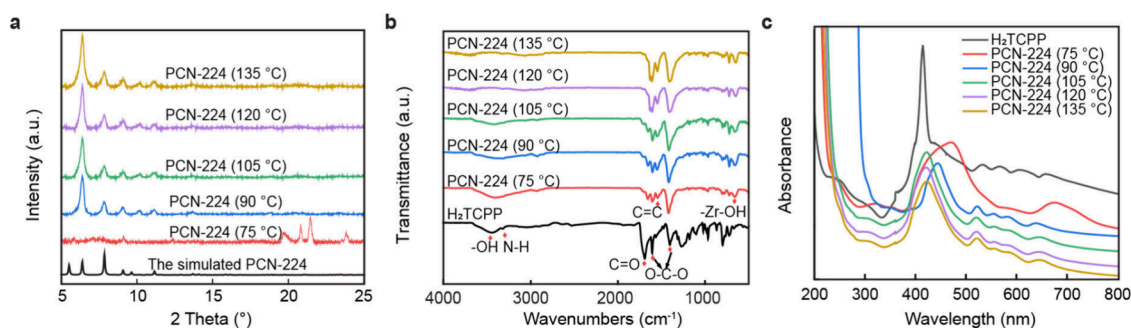


Figure 2. Physicochemical behaviors of the PCN-224 nanoparticles at different solvothermal synthesis temperatures. (a) PXRD pattern, (b) FT-IR spectra, and (c) UV-vis absorption spectra of H₂TCPP, PCN-224 (75 °C), PCN-224 (90 °C), PCN-224 (105 °C), PCN-224 (120 °C), and PCN-224 (135 °C) nanoparticles.

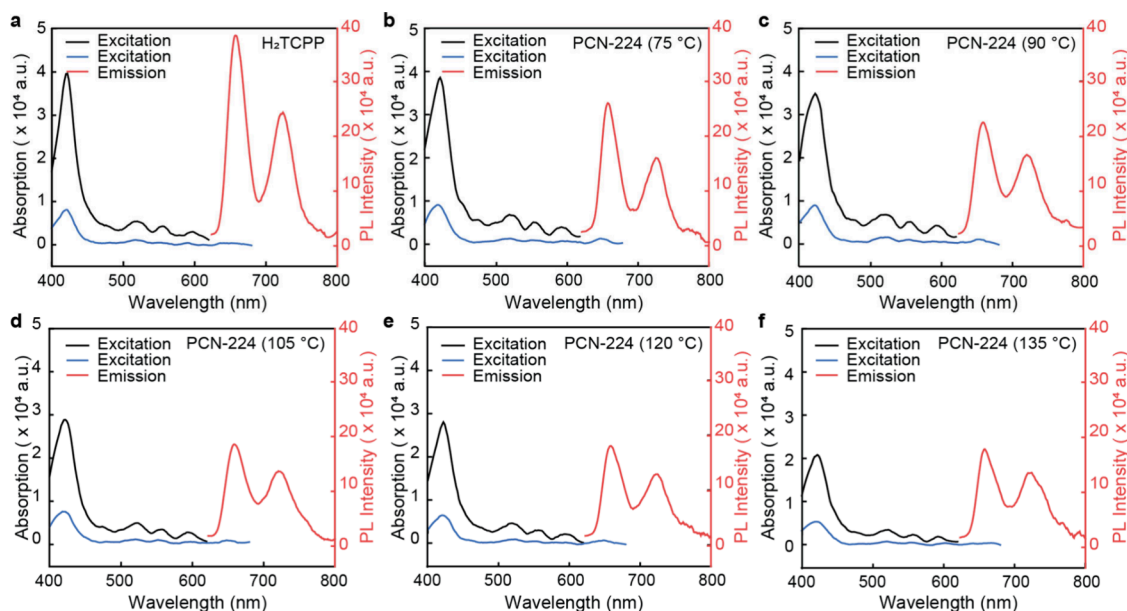


Figure 3. Photoluminescence properties of the H₂TCPP ligand and the PCN-224 nanoparticles at different solvothermal synthesis temperatures. Adsorption and PL intensity of (a) H₂TCPP, (b) PCN-224 (75 °C), (c) PCN-224 (90 °C), (d) PCN-224 (105 °C), (e) PCN-224 (120 °C), and (f) PCN-224 (135 °C). Curve assignments: λ_{em} = 680 nm (black curve); λ_{em} = 720 nm (blue curve); λ_{ex} = 420 nm (red curve).

°C), 60.3 ± 5.8 nm (120 °C), and 57.7 ± 11.8 nm (135 °C), respectively. At the solvothermal temperature increased from 90 to 135 °C, the average nanoparticle size of the PCN-224 nanoparticles showed a gradual increasing tendency, while the standard deviation significantly expanded approximately 3-fold its initial value. The expanded standard deviation indicated that as the temperature increased, the nanoparticle size distribution of PCN-224 became less uniform. In Figure 1g, we selected PCN-224 (105 °C) nanoparticles with relatively uniform size for energy dispersive spectroscopy (EDS) mapping analysis, confirming the existence of C, N, O, and Zr elements of the PCN-224 nanoparticles.

3.2. Physicochemical Study of PCN-224 Nanoparticles

In Figure 2, we studied the evolution of crystallinity and physicochemical behaviors of the PCN-224 nanoparticles depending on precise solvothermal synthesis temperatures. Figure 2a depicts the PXRD pattern of the PCN-224 nanoparticles at different solvothermal synthesis temperatures to compare with the crystallinity structure of simulated PCN-224. The pattern of PCN-224 (75 °C) nanoparticles, with the main diffraction peaks at 9.1, 12.4, 19.7, 20.8, 21.5, 23.5, and 23.8°, showed an obvious mismatch with the simulated PCN-

224 pattern, attributed to insufficient crystallization caused by the lower temperature. As the solvothermal synthesis temperature exceeds 75 °C, the different peaks of PCN-224 (90–135 °C) nanoparticles at 6.4, 7.9, 9.1, and 11.2° represent the crystal planes of [022], [222], [004], and [224], respectively, consistent with the simulated PCN-224.²⁷ In addition, the diffraction peak of synthesized PCN-224 (75–135 °C) nanoparticles was widened after treatment in aqueous solution for 1 month but still maintained a relatively complete crystal structure (Supporting Information Figure S1). Thus, the crystal structure of the PCN-224 nanoparticles was confirmed by PXRD, indicating that the structure can be completely formed above 75 °C and remained stable.

Figure 2b displays the FT-IR spectra of the H₂TCPP ligand and the PCN-224 nanoparticles. For H₂TCPP spectra, the characteristic peak around 3440 cm⁻¹ corresponds to the -OH stretching vibration and the bands at 3315 cm⁻¹ and 966 cm⁻¹ are assigned to N-H vibrations.²⁸ In addition, the peaks situated at 1685 and 1540 cm⁻¹ can be attributed to C=O and C-C stretching bonds, respectively. The features appearing at 1600 and 1405 cm⁻¹ were ascribed to the characteristic vibration of the -O-C-O- bond.²⁹ Notably, the C=O

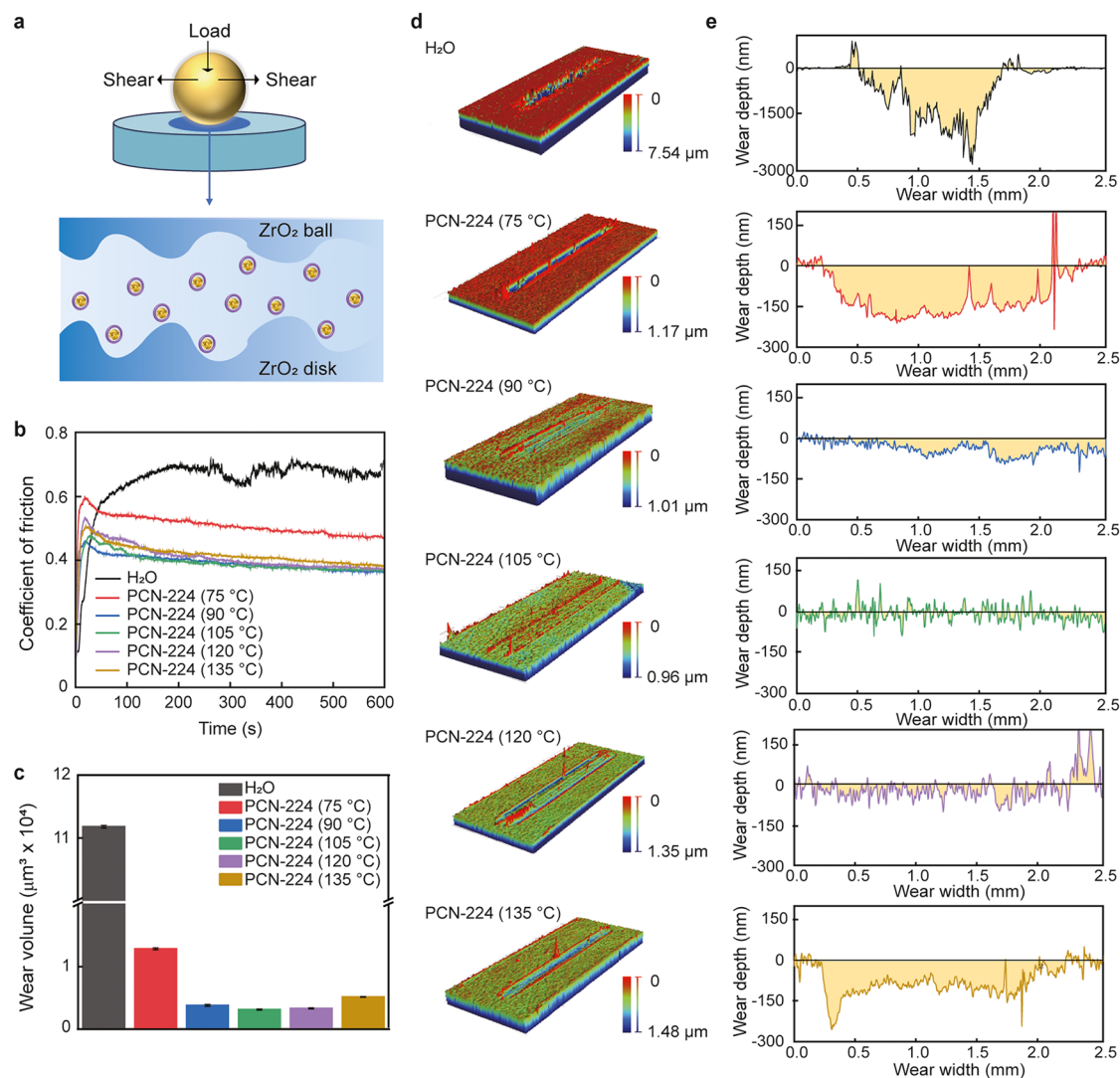


Figure 4. Aqueous lubrication performance of PCN-224 nanoparticles at different solvothermal synthesis temperatures. (a) Scheme of tribological experiment. (b) Friction curves, (c) wear volume, (d) 3D morphological profiles, and (e) wear depth of H₂O, PCN-224 (75 °C), PCN-224 (90 °C), PCN-224 (105 °C), PCN-224 (120 °C), and PCN-224 (135 °C) lubricating additives in water (load, 10 N; frequency, 2 Hz; temperature, 25 °C; concentration, 10 mg/mL).

stretching vibration peak at 1685 cm⁻¹ disappeared in the FT-IR spectra of all PCN-224 nanoparticles, indicating alterations in the carboxyl groups of H₂TCPP. The characteristic absorption peaks at 658 cm⁻¹ corresponding to -Zr-OH further confirmed the successful synthesis of PCN-224 nanoparticles. Compared to H₂TCPP, the peak intensity of PCN-224 (75–135 °C) nanoparticles was reduced, attributed to the binding of Zr clusters. The existence of -OH, N-H, and C-O groups demonstrated the hydrophilicity properties of all synthesized PCN-224 (75–135 °C). The TG analysis of PCN-224 (105 °C) showed that framework decomposition occurred above approximately 450 °C (Figure S2), exhibiting good thermal stability. The FT-IR analysis indicated that the coordination between ligand and metal can be achieved even at lower temperature.

UV-vis was performed to determine the optical properties of the PCN-224 nanoparticles. In Figure 2c, the spectrum of the H₂TCPP ligand displayed a typical Soret band at 420 nm and four Q bands at 532, 569, 600, and 660 nm, respectively.³⁰ In contrast, the spectrum of PCN-224 (75 °C) nanoparticles

exhibited a significantly broadened Soret band and poorly observed Q bands. The spectral broadening observed for PCN-224 (75 °C) nanoparticles can be attributed to a broader particle size distribution and the presence of crystalline defects.³¹ In addition, the spectra of the other PCN-224 (90–135 °C) nanoparticles showed well-defined Soret at 420 nm and four Q-bands at 518, 554, 585, and 645 nm, respectively. Compared to the free H₂TCPP ligand, the shift of the Q bands confirmed the successful integration of porphyrin ligands into the MOF framework without central metallization.³² Therefore, although all synthesized nanoparticles can form the PCN framework through coordination, PCN-224 nanoparticles with clear crystal structures and optical properties can be constructed only when the solvothermal synthesis temperature rises to 90 °C or above.

3.3. Photoluminescence Study of PCN-224 Nanoparticles

Excitation and emission spectra can be used to assess the recombination efficiency of photogenerated electron-hole pairs.³³ Figure 3 shows the photoluminescence properties of free H₂TCPP ligand and the PCN-224 nanoparticles. In Figure

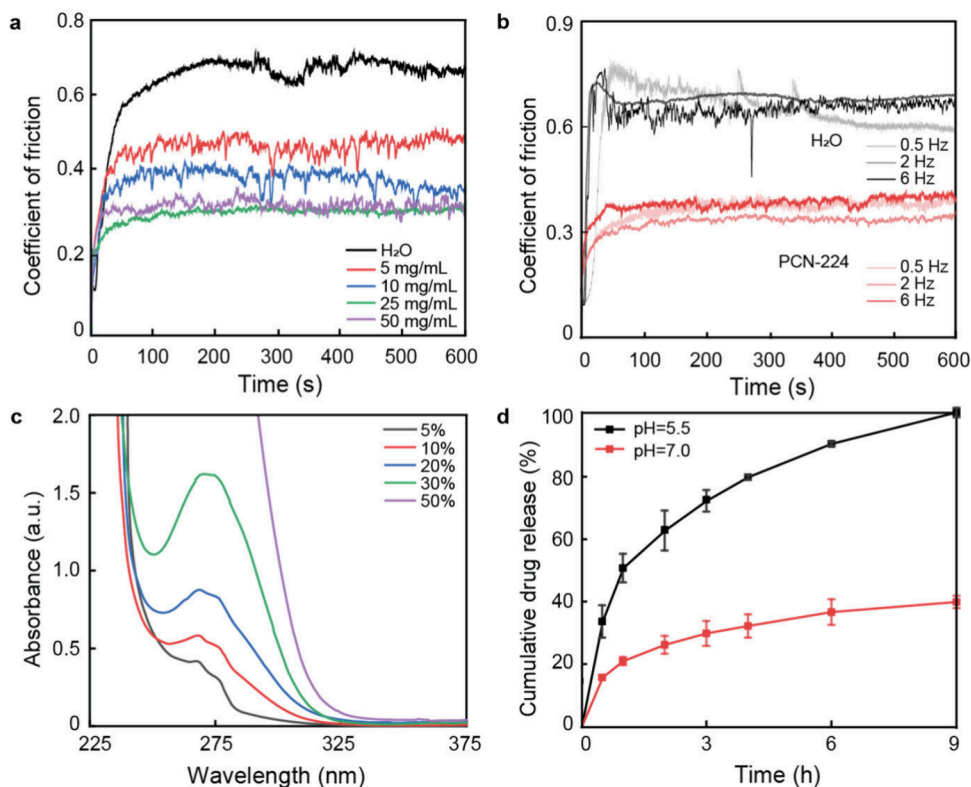


Figure 5. Optimized aqueous additives of PCN-224 as OA treatment nanocarriers. (a, b) Friction curves at different concentration and frequency of pure water and PCN-224 (105 °C) additives. (c) UV–vis spectra of the supernatant of PCN-224 (105 °C) suspension after drug encapsulation. (d) Cumulative drug release of DS-loaded PCN-224 (105 °C) nanocarriers in H₂O at pH = 5.0 and 7.0.

3a, under 420 nm excitation, both the free H₂TCPP ligand and the PCN-224 nanoparticles exhibited dual PL emission bands at 680 and 720 nm (red curves). This phenomenon suggested that the Zr–porphyrin MOFs can be used as densely packed molecular emitters integrated onto plasmonic nanoparticles. In terms of excitation wavelength specificity, the 680 nm emission band was more prominent than the 720 nm band for all materials. In Figure 3b–f, this emission profile corresponded directly to the characteristic band absorptions observed in the UV–vis spectra (black and blue curves), confirming that the radiative recombination originates from the porphyrin chromophores.^{34,35} Critically, the absence of spectral shifts or quenching in the emission band of PCN-224 nanoparticles indicated that the free-base porphyrin units retained their electronic structure upon incorporation into the metal–organic framework without metallization of the porphyrin units. This successful immobilization of the intact photoactive linker is essential for the intended photophysical applications of the PCN-224 nanoparticles.

The PL effect of H₂TCPP was derived from the π – π electron radiation transition of the porphyrin ring conjugated system. Upon coordination of the ligand with metal nodes to construct the crystalline skeleton, a distinct quenching effect was induced, resulting in the PL intensity of PCN-224 being lower than that of H₂TCPP. To be specific, the PL intensity of PCN-224 (75 °C) nanoparticles displayed a relatively lower intensity than that of the H₂TCPP ligand but higher than that of PCN-224 (90–135 °C) nanoparticles. Based on the TEM image and XRD pattern, it can be inferred that the higher PL intensity of PCN-224 (75 °C) nanoparticles can be attributed to the reduced degree of tight packing between units by the free porphyrin molecules that did not fully participate in

coordination, partially alleviating aggregation quenching. In contrast, the PL intensity of PCN-224 (90–135 °C) nanoparticles significantly decreased with increasing solvothermal synthesis temperatures. This trend was related to more complete H₂TCPP ligands and Zr cluster coordination, facilitating more efficient charge transfer and nonradiative energy relaxation pathways, thereby leading to a further reduction in PL intensity. Therefore, the prominent 680 nm red emission peak of the PCN-224 nanoparticles confirms its potential as a red fluorescent nanocarrier for cell imaging applications, and the stable PL intensity offers reproducibility advantages for biological imaging.

3.4. Tribological Study of PCN-224 Nanoparticles

To systematically explore the lubrication performance of the PCN-224 nanoparticles synthesized at different solvothermal temperatures, we constructed a friction test under a ball-on-disc contact mode (Figure 4). Zirconia ceramics are chosen as friction pairs, which are typical ball head materials for total hip arthroplasty for treatment of OA (Figure 4a).³⁶ The COF is collected by UMT Tribo machine and recorded in real time via a computer system connected to the testing machine. First, we measured the COF of pure water under 10 N load, which quickly rose and stabilized at 0.68 after running 600 s (Figure 4b). Also, the COF of PCN-224 (75 °C) nanoparticles in water showed a quick rise to 0.60 and then reduced to approximately 0.53. The average COF of other additives were reduced in the whole reciprocating process to be around 0.36 for PCN-224 (90 °C), 0.36 for PCN-224 (105 °C), 0.37 for PCN-224 (120 °C), and 0.39 for PCN-224 (135 °C), respectively. Compared with the pure water and PCN-224

(75 °C), the antifriction ability of the PCN-224 (90–135 °C) lubricating nanocarriers was notably enhanced.

Panels c–e of Figure 4 show the 3D wear scar profile and average wear volume of pure water and the PCN-224 nanocarriers as aqueous lubrication additives. For pure water, wear volume on ZrO₂ disk was $11.17 \times 10^4 \mu\text{m}^3$. By adding the PCN-224 nanoparticles, the wear volume of the disk decreased to $1.29 \times 10^4 \mu\text{m}^3$ for PCN-224 (75 °C), $0.38 \times 10^4 \mu\text{m}^3$ for PCN-224 (90 °C), $0.31 \times 10^4 \mu\text{m}^3$ for PCN-224 (105 °C), $0.33 \times 10^4 \mu\text{m}^3$ for PCN-224 (120 °C), and $0.52 \times 10^4 \mu\text{m}^3$ for PCN-224 (135 °C), respectively. Compared to the pure water and PCN-224 (75 °C) additives, the superior antifriction and antiwear performance of PCN-224 (90–135 °C) lubricating additives can be attributed to its more complete crystal structure and uniform spherical morphology. The complete framework structure provided enhanced mechanical stability under shear, while the uniform spherical nanoparticles can promote rolling friction at the interface. In contrast, PCN-224 nanoparticles synthesized at 75 °C showed lower crystallinity and spindle-like morphology that compromised the friction-reducing effectiveness and load-bearing capacity.

We further studied the morphological contours of the worn surfaces based on different lubricants of pure water and PCN-224 lubricating additives to understand the antiwear ability of the prepared samples (Figure 4d,e). The wear scar of pure water exhibited significant length (1.87 mm) and depth (2.76 μm), accompanied by a highly rough surface (7.54 μm). Compared with pure water, the 3D optical images of the worn surface of PCN-224 lubricating additives exhibited being narrower and shallower than those of pure water. The wear depth decreased to 0.20 μm for PCN-224 (75 °C), 0.09 μm for PCN-224 (90 °C), 0.05 μm for PCN-224 (105 °C), 0.06 μm for PCN-224 (120 °C), and 0.18 μm for PCN-224 (135 °C) additives, respectively. By making comparison with pure water, we found that almost 52% reduction for both friction coefficient and wear volume were achieved by using PCN-224 as aqueous lubricating additives. Among the synthesized nanoparticles, the worn surface of PCN-224 (105 °C) lubricating additives appeared to have improved smoothness with shallower scratches. These results indicated that PCN-224 (105 °C) as aqueous additives can significantly reduce both the friction coefficient and wear volume.

3.5. Optimization of PCN-224 Nanocarriers for Drug Delivery of OA Treatment

To optimize PCN-224 nanoparticles as bifunctional OA treatment nanocarriers, we further conducted both lubrication and drug delivery performance studies (Figure 5). Tribological study in aqueous medium was examined at various concentrations and sliding frequencies. In Figure 5a, pure water experienced significant lubrication failure at 45 s, accompanied by a violent seizure phenomenon between the friction pairs, and the COF rose rapidly to 0.63. With the rising concentration of PCN-224 (105 °C) additives, COF decreased to 0.47 (5 mg/mL), 0.38 (10 mg/mL), 0.32 (25 mg/mL), and 0.33 (50 mg/mL), respectively. The introduction of PCN-224 additives can effectively improve the lubrication performance of pure water, and the best addition concentration of the PCN-224 additives for aqueous lubrication was 25 mg/mL. Furthermore, we conducted frequency-dependent tribological performance of the PCN-224 nanoparticles at the optimized concentration. In Figure 5b, the average COF of pure water remained a high value in the whole test, reaching 0.61 (0.5

Hz), 0.67 (2 Hz), and 0.65 (6 Hz), respectively. After adding the additives, the average COF of PCN-224 reduced to about 0.35 (0.5 Hz), 0.33 (2 Hz), and 0.40 (6 Hz), respectively. These results indicated the enhanced lubricating performance and the effective frequency stability of optimized PCN-224 (105 °C) additives.

Due to the excellent biocompatibility of the PCN-224 nanoparticles,³⁷ we further studied drug delivery performance of the optimized superior lubricating nanocarriers for OA treatment. DS was selected as a model drug for OA treatment because of its good anti-inflammatory capability.³⁸ First, we studied UV–vis spectra of DS in pure water at different concentrations to build the calibration curve, and the strong absorbance at $\lambda = 276 \text{ nm}$ was used to evaluate the drug loading and releasing properties of the PCN-224 (105 °C) nanocarriers (Figure S3). Then, PCN-224 nanocarriers were mixed with different concentrations of DS for drug loading, and the solutions were centrifuged after 24 h. We conducted UV–vis spectra of the supernatants to study the loading capacity of PCN-224@DS nanocarriers (Figure 5c). For the 5% of DS mixed with PCN-224 nanocarrier, the absorbance of DS at 276 nm almost disappeared, indicating that the drugs were fully encapsulated into the PCN-224 nanocarriers. Based on the standard curve of DS, the drug loading capacity (DLC) and drug loading efficiency (DLE) of different amounts of DS encapsulation in the PCN-224 nanocarriers were evaluated in Table S1. The concentration of 20% DS for the PCN-224 nanocarriers showed the comprehensive promising DLC (19.4%) and DLE (97.2%), which was then ultimately chosen as the optimized concentration.

In addition, we conducted drug release behavior from the PCN-224@DS nanocarriers at different pH values. The pH value of normal synovial fluid in the joint cavity is about 7.0, whereas those of inflammatory synovial fluid are about 5.5.³⁹ The absorbance of DS was significantly higher at acidic (pH 5.0) environment than that of the physiological environment (pH 7.4) via the time-dependent UV–vis spectra (Figure S4). At pH 7.4, the DS release from the PCN-224@DS nanocarriers was about 18% within 0.5 h, while the cumulative release of DS at pH = 5.5 increased to 35% (Figure 5d). After 9 h, nearly 100% of DS was released from the nanocarriers at pH = 5.5, while only 39.8% DS was released at pH = 7.0, which can be attributed to cleavage of an acid-labile linker of PCN-224. Therefore, optimized PCN-224 (105 °C) nanocarriers with great antifriction performances and acid-responsive drug release abilities can effectively serve as OA treatment nanocarriers.

4. CONCLUSION

In conclusion, we successfully synthesized a series of size-tunable PCN-224 nanoparticles under a precise solvothermal temperature gradient ranging from 75 to 135 °C, establishing the structure–activity relationship between synthesis temperatures and photoluminescence properties of PCN-224 as bifunctional OA treatment nanocarriers. As the solvothermal synthesis temperature increased, the morphology of PCN-224 nanoparticles gradually transitioned from spindle-shaped to spherical-shaped structures and their average size became more uniform (from 229.3 ± 25.6 to $57.7 \pm 11.8 \text{ nm}$). The photoluminescence intensity of PCN-224 decreased as the solvothermal synthesis temperature increased. Friction tests demonstrated that almost 52% reduction for both friction coefficient and wear volume were realized using PCN-224

nanoparticles as aqueous lubricating additives, which was better than that of pure water. Among the synthesized nanoparticles, PCN-224 (105 °C), as efficient drug nanocarriers, can encapsulate 19.4% drugs, achieving excellent antifriction performance and acid-responsive drug release behavior for synergistic OA therapy. Therefore, our work not only provided deep insight for understanding of the precise temperature-gradient-modulated structure of PCN-224 but also offered valuable design strategies for the rational fabrication of MOFs-based bifunctional nanocarriers in the field of biomedical fields.

■ ASSOCIATED CONTENT

Data Availability Statement

Supporting data for this work can be accessed through the [Supporting Information](#) file or from the authors upon reasonable request.

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/prechem.5c00448>.

(Figure S1) Water stability of PCN-224 nanoparticles; (Figure S2) thermal stability analysis of PCN-224 nanoparticles; (Figure S3) calibration curve of DS in pure water; (Table S1) nanocarriers' drug loading capacity and efficiency; (Figure S4) in vitro DS release of nanocarriers at different pHs (PDF)

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Notes

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